

Numpy Notes

Numpy

- Numpy is a python library
- Numpy is used for working with arrays.
- Numpy is short for "numerical python"

What is Numpy?

Numpy is a python library used for working with arrays. It also has functions for working in domain of linear algebra, fourier transform, and matrices. Numpy was created in 2005 by Travis Oliphant. It is an open source project and you can use it freely.

Why use Numpy?

In python we have lists that serve the purpose of arrays, but they are slow to process.

Numpy aims to provide an array object that is up to 50x faster than traditional python lists.

The array object in Numpy is called ndarray. It provides a lot of supporting functions that make working with ndarray very easy.

Arrays are very frequently used in data science where speed and resources are very important.

Why is Numpy faster than lists?

Numpy arrays are stored at once continuous place in memory unlike lists, so processes can access and manipulate them very efficiently.

This behavior is called locality of reference in computer science.

This is the main reason why Numpy is faster than lists. Also it is optimized to work with latest CPU architectures.

Which language is Numpy written in?

- Numpy is a python library and is written partially in python, but most of the parts that requires fast computation are written in C or C++.

Installation of numpy

If you have python and pip already installed on a system, then installation of numpy is very easy.
Install it using command:

```
c:\users\Your Name\pip install numpy
```

import numpy

```
import numpy
```

Ex:

```
import numpy
```

```
arr = numpy.array([1, 2, 3, 4, 5])
```

```
print(arr)
```

numpy as np

numpy is usually imported under the np alias.

Create an alias with the as keyword while importing.

```
import numpy as np
```

Now the Numpy package can be referred to as np and instead of numpy.

Ex:

```
import numpy as np
```

```
arr = np.array([1, 2, 3, 4, 5])
```

```
print(arr)
```

checking numpy version

The version string is stored under `--version--` attribute.

```
import numpy as np
```

```
print(np.__version__)
```


Create a Numpy ndarray object

Numpy is used to work with arrays. The array object in Numpy is called ndarray. We can create a Numpy ndarray object by using the `array()` function.

Ex:

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print(arr)
print(type(arr))
```

To create an ndarray, we can pass a list, tuple, or any array-like object into the `array()` method, and it will be converted into an ndarray.

Ex:

use a tuple to create a Numpy arrays

```
import numpy as np
arr = np.array((1, 2, 3, 4, 5))
print(arr)
```

Dimensions in Arrays

A dimension in arrays is one level of array depth (nested arrays).

nested arrays are arrays that have arrays as their elements.

0-D Arrays

0-D arrays, or scalars, are the elements in an array. Each value in an array is a 0-D array.

Ex:

create a 0-D array with value 42

```
import numpy as np
arr = np.array(42)
print(arr)
```

1-D Arrays

An array that has 0-D arrays as its elements is called unidimensional or 1-D array.

These are the most common and basic arrays.

create a 1-D array containing the values 1, 2, 3, 4, 5:

```
import numpy as np  
arr = np.array([1, 2, 3, 4, 5])  
print(arr)
```

2-D Arrays

An array that has 1-D arrays as its elements is called 2D array.

These are often used to represent matrix or 2nd order tensors.

Ex1

create a 2-D array containing two arrays, with the values 1, 2, 3 and 4, 5, 6:

```
import numpy as np  
arr = np.array([1, 2, 3], [4, 5, 6])  
print(arr)
```

3-D Arrays

An array that has 2-D arrays (matrices) as its elements is called 3D array.

These are often used to represent a 3rd order tensor.

Ex1

create a 3-D array with two 2-D arrays, both containing two arrays with values 1, 2, 3 and 4, 5, 6:

```
import numpy as np  
arr = np.array([ [1, 2, 3], [4, 5, 6] ], [ [1, 2, 3], [4, 5, 6] ])   
print(arr)
```


check number of dimensions?

Numpy Arrays provides the `ndim` attribute that returns an integer that tells us how many dimensions the array have.

ex1

check how many dimensions the arrays have;

```
import numpy as np
```

```
a = np.array([42])
```

```
b = np.array([1, 2, 3, 4, 5])
```

```
c = np.array([[1, 2, 3], [4, 5, 6]])
```

```
d = np.array([[[1, 2, 3], [4, 5, 6]], [[1, 2, 3], [4, 5, 6]]])
```

```
print(a.ndim)
```

```
print(b.ndim)
```

```
print(c.ndim)
```

```
print(d.ndim)
```

High Dimensional Arrays

An array can have any number of dimensions. When the array is created, you can define the number of dimensions by using the `ndim` argument.

ex2

create an array with 5 dimensions and verify that it has 5 dimensions.

```
import numpy as np
```

```
arr = np.array([1, 2, 3, 4], ndim=5)
```

```
print(arr)
```

```
print('number of dimensions: ', arr.ndim)
```

In this array, the innermost dimension (0th dim) has 4 elements, the 4th dim has 1 element that is the vector, 3rd dim has 1 element that is the matrix with vector, the 2nd dim has 1 element that is 3D array and 1st dim has 1 element that is an array.